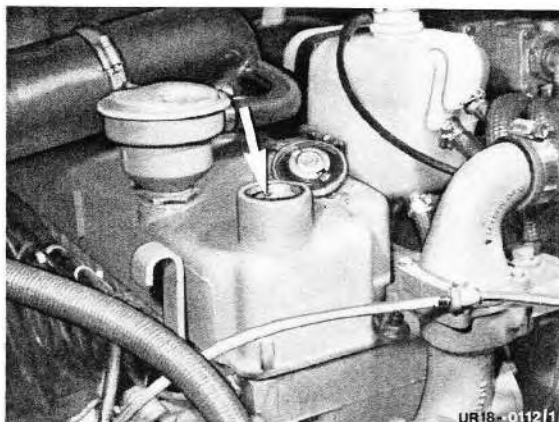
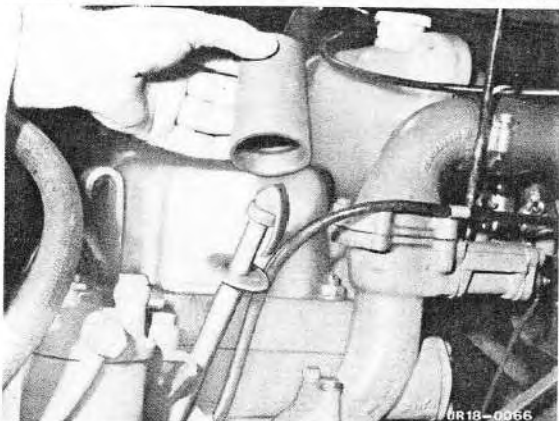


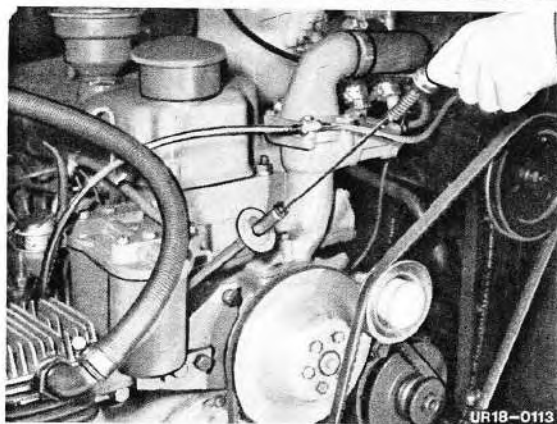


10 Fill system with engine oil, referring to 01.8-1.3/2 for quantity.

11 Run engine for brief period.



12 Detach protective cap from dipstick.



13 Lift up locking lever on dipstick and check level on dipstick, topping up with engine oil if necessary.

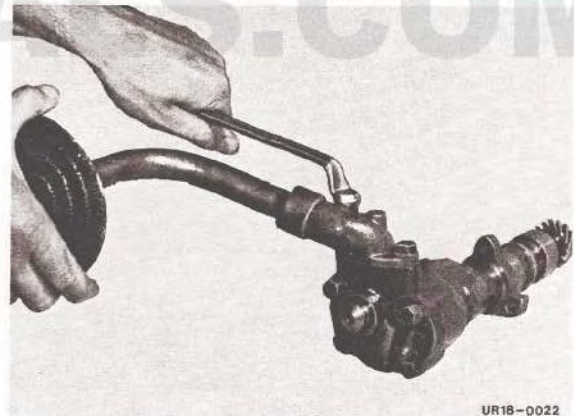
Removal

- 1 Remove oil pan, referring to 2.1/1.
- 2 Remove oil pump fastening bolt and take oil pump out of crankcase.
- 3 For blocking oil pump, detach screw plug at crankcase outside and release oil pump using a soft drift.
- 4 Detach suction pipe from oil pump and clean pump.
- 5 Disassemble oil pump, referring to 3.2/1.



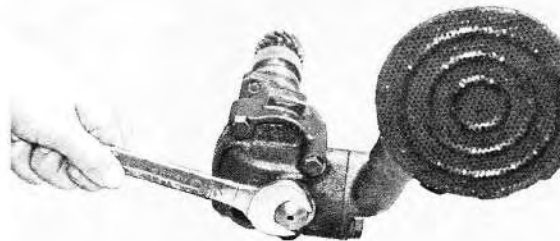
Installation

- 1 Attach suction pipe with new gasket.
- 2 Introduce oil pump into crankcase and tighten. See 1.2/1 for tightening torque.
- 3 Install oil pan, referring to 2.1/1.



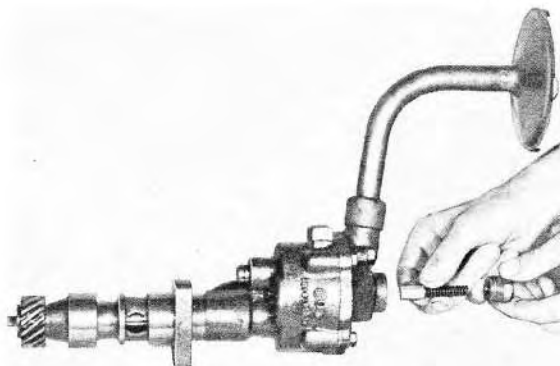
Disassembly

- 1 Remove oil pump, referring to 3.1/1.
- 2 Unscrew and remove screw plug from oil delivery valve.



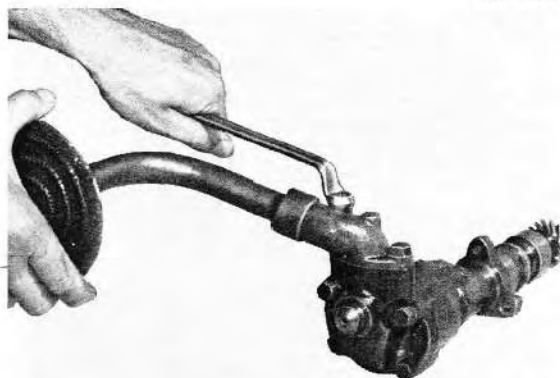
UR18-0019

- 3 Remove compression spring and piston, checking for wear.



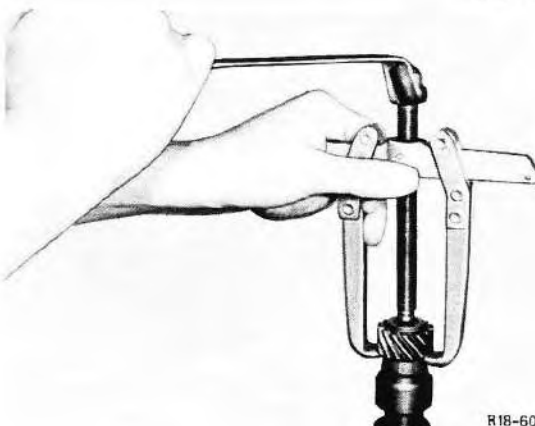
UR18-0023

- 4 Unscrew suction pipe from cover.
- 5 Clamp oil pump in soft supports.
- 6 Detach helical gear.
- 7 Unscrew oil pump cover.
- 8 Take oil pump gear and drive shaft out of oil pump body.
- 9 Force oil pump gear off drive shaft.

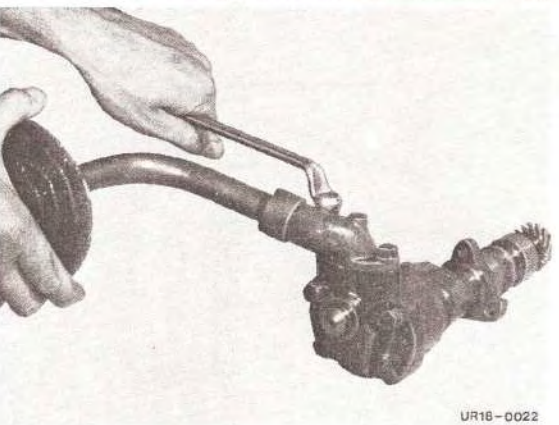
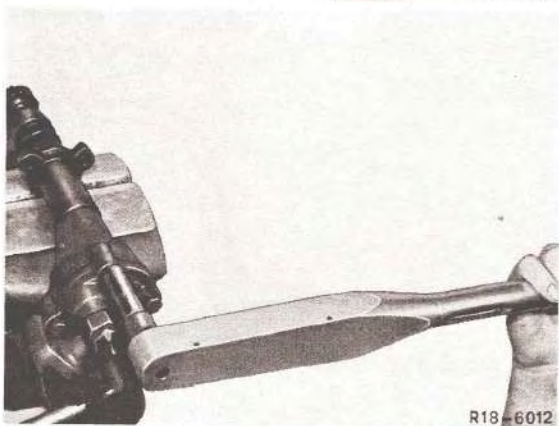
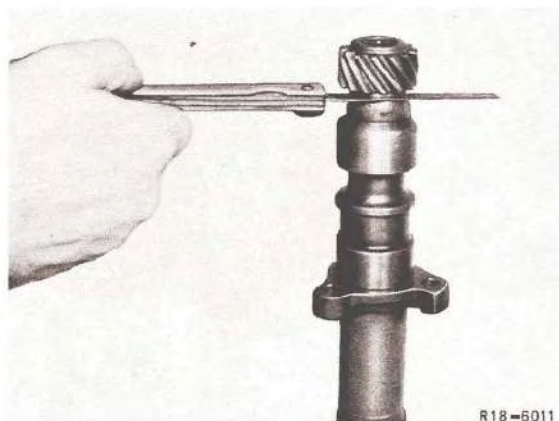


UR18-0022

- 10 Take oil pump gear off oil pump shaft.
- 11 Force oil pump shaft out of body.



R18-6026



Assembly

- 1 Heat driving oil pump gear to approx. 80°C and fit on drive shaft.
- 2 Introduce drive shaft and gear into body.
- 3 Heat helical gear to approx. 80°C and fit on drive shaft.
- 4 Check end play of drive shaft, referring to 1.2/1.
- 5 Force oil pump shaft and gear into body.
- 6 Check end play of oil pump gear, referring to 1.2/1.
- 7 Bolt cover to oil pump body, referring to 1.2/1 for tightening torque.
- 8 Introduce piston and compression spring into oil pump cover. Insert thrust screw with fixing agent No. 2 and tighten. See 1.2/1 for tightening torque.
- 9 Attach strainer to cover with new gasket. See 1.2/1 for tightening torque.
- 10 Install oil pump, referring to 3.1/1.

Removal

Note: Sequence of operations is identical for all engine models, mounted for engine model 353.959 up to engine No. 528 529.

- 1 Raise cab,
referring to 60.5–2.1/1 or 60.5–2.2/1.
- 2 Drain off coolant at engine.

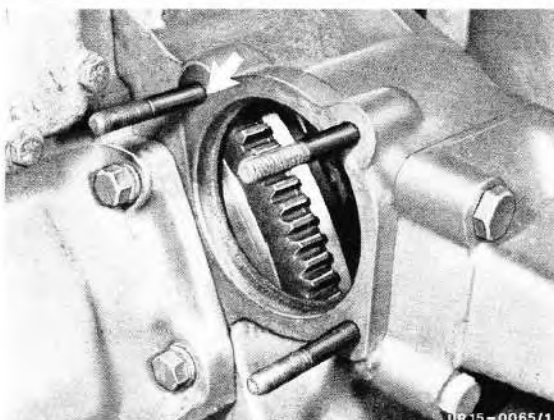
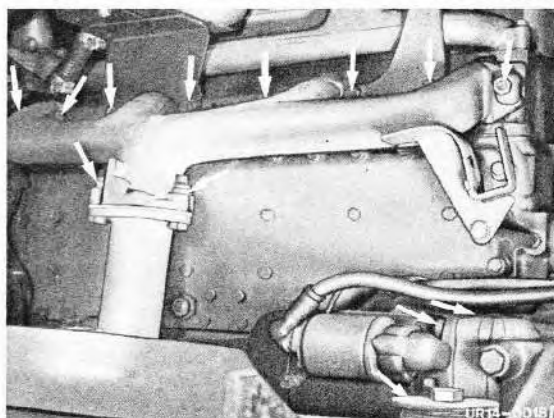
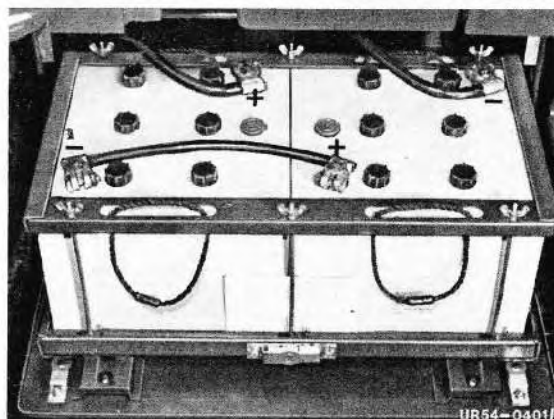
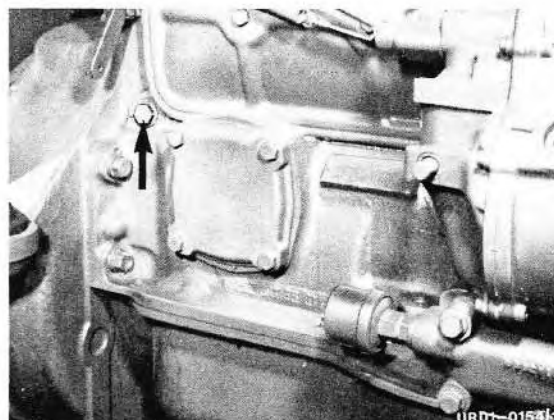
- 3 Disconnect battery.

Note: First disconnect (–) and then (+) terminal.

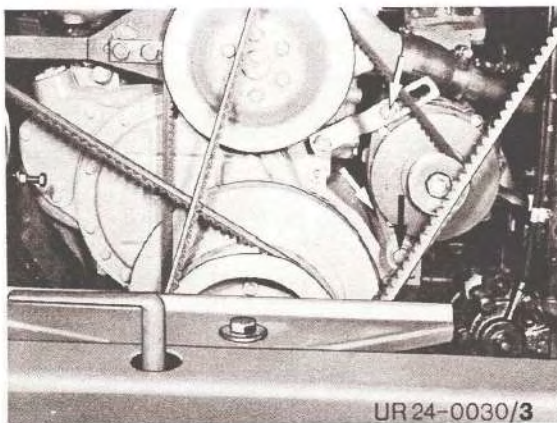
- 4 Detach exhaust manifold and starter.

Note: For pressure charged engine, detach turbo-charger.
Referring to 09.13–2.1/1.

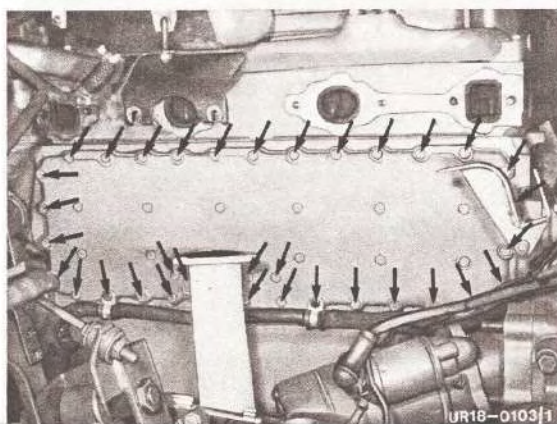
- 5 Remove inner starter fastening stud.



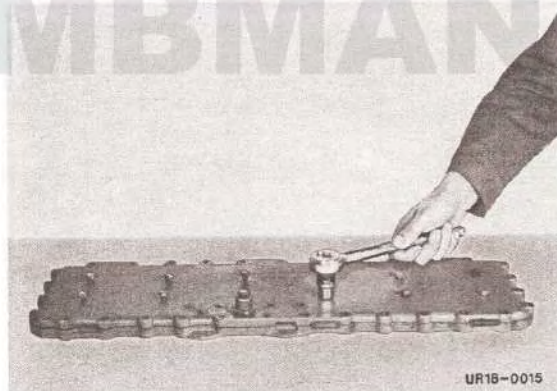
352/352 A



- 6 Detach generator, but do **not** disconnect wiring.

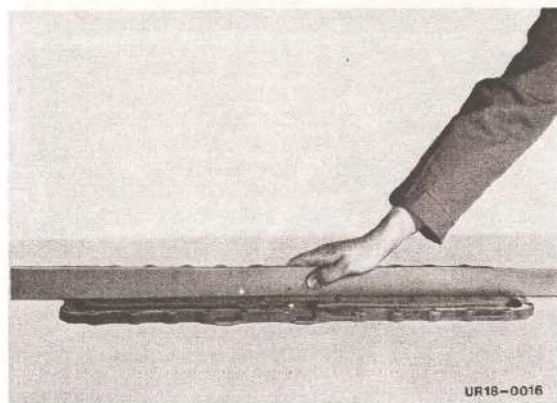


- 7 Remove outer fastening bolts from oil cooler as well as 2 bolts at left and right-hand sides of oil pressure relief valve. Remove oil cooler.



- 8 Unscrew oil cooler cover from oil cooler housing.

- 9 Detach oil pressure relief valve in oil cooler, clean and check, exchanging if necessary. See table 1.1/1 for oil pressure relief valve.



- 10 Clean oil cooler, checking sealing surfaces for distortion.

Fitting

Note: Sequence of operation is identical for all engine models. Fitted on engine model 353,959 up to engine end No. 528,529.

1 Thoroughly clean the sealing surfaces between crankcase and oil cooler as well as between the two plates of the oil cooler.

2 Fit four paper seals to the drillings indicated with arrows and lay the gasket on the oil cooler housing sealing surface.

3 Fit oil cooler to the oil cooler body, screw in bolts with washers, do not tighten.

4 Fit oil cooler with new gasket to the crankcase and tighten all bolts working diagonally from outside inwards.

Note: For easier fitting screw two studs into the crankcase.

5 Fit oil pressure relief valve and tighten. Tightening torque, see 1.2/1.

6 Fit inner anchor bolts for fixing starter. Tightening torque, see 1.2/1.

7 Fit starter with sealant No. 3, screw on nuts with sealant No. 4 and tighten. Tightening torque, see 1.2/1.

8 Fit exhaust manifold with gasket and tighten. Tightening torque, see 1.2/1.

Note: With turbocharged engine fit turbocharger, 09.13.-2,1/1.

9 Fit generator and Vee belt. Tension Vee belt, see 13.14-2,6/1.

10 Connect battery.

Note: First connect (+), then (–) terminal.

11 Fill cooling circuit to capacity, see 01.8-1,3/2.

Note: The expansion tank should be up to 2/3 full.

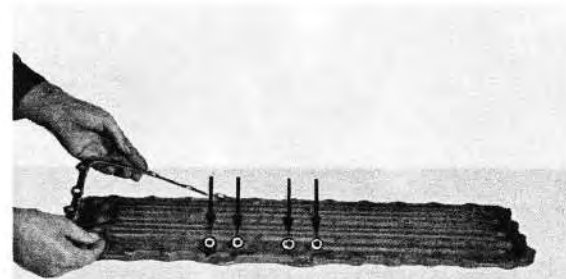
12 Bleed engine, see 5,1/1.

13 Ensure correct engine oil level.

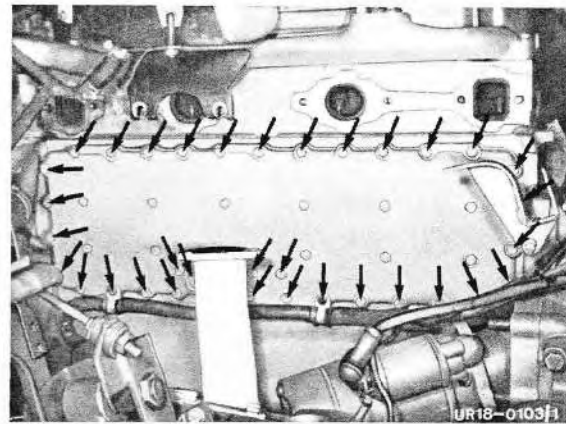
14 Turn engine with starter, but the engine must **not** fire. When the gauge shows oil pressure, the engine can be started.

15 Test oil cooler for leaks, check the engine oil level again.

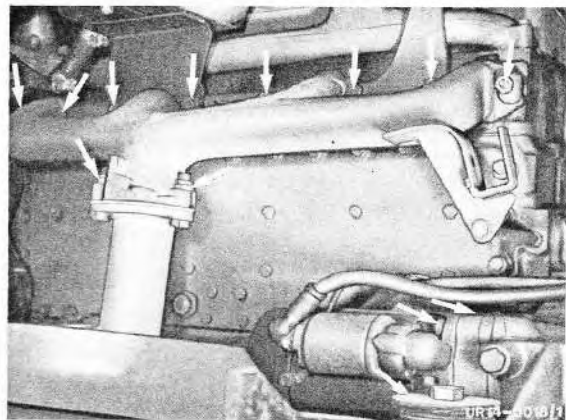
16 Lower cab, see 60.5-2,1/1 or 2,2/1.



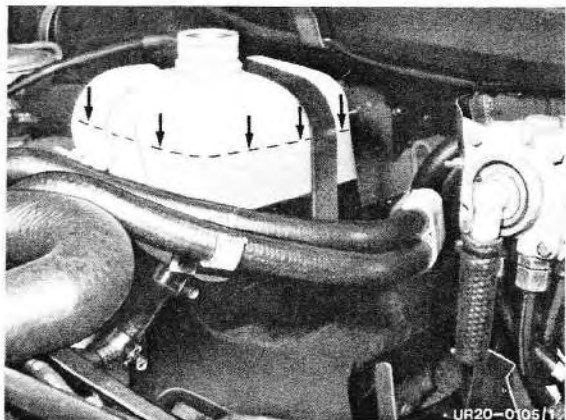
UR18-0014



UR18-0103/1



UR18-0015/1



UR20-0105/1

Removal**From engine No. 528 530**

Note: Sequence of operations is identical for all engine models.

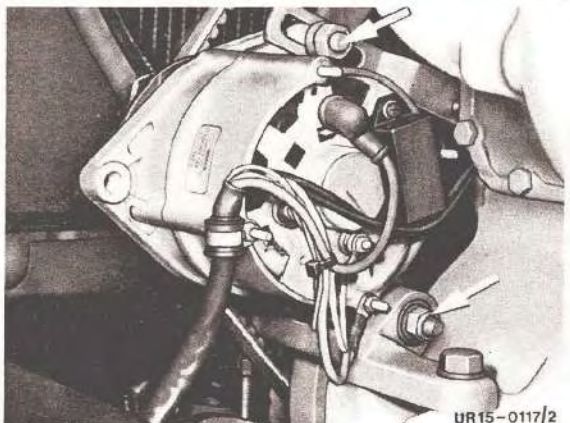
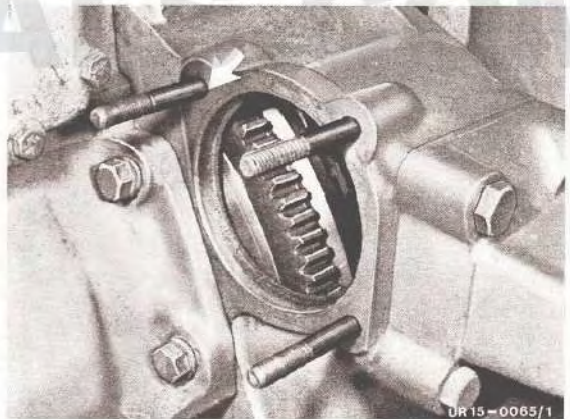
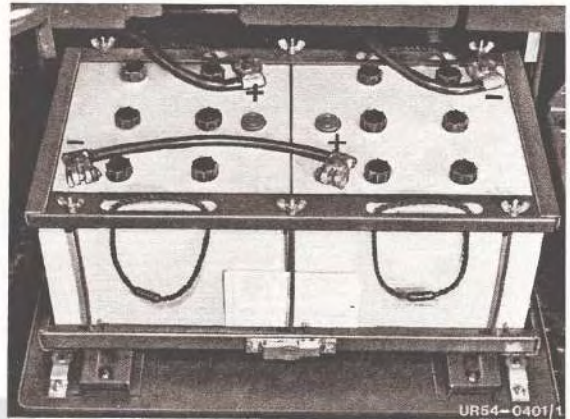
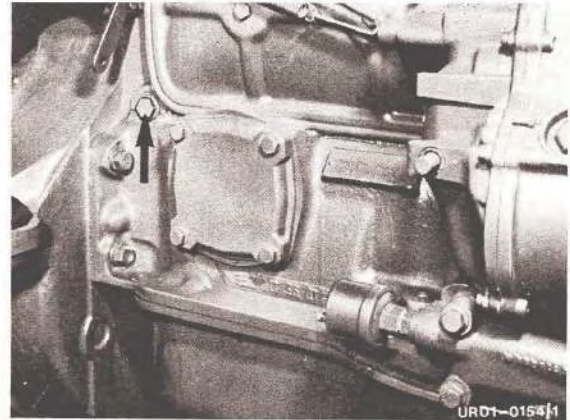
- 1 Raise cab,
referring to 60.5–2.1/1 or 60.5–2.2/1.
- 2 Drain off coolant at engine.

- 3 Disconnect battery.

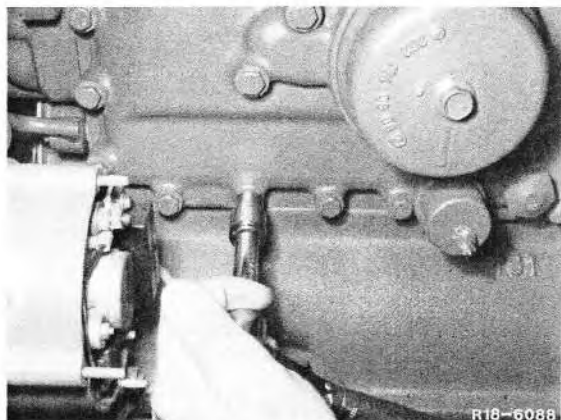
Note: First connect (–) and then (+) terminal.

- 4 Remove turbocharger, referring to 09.13–2.1/1.
- 5 Detach exhaust manifold and starter, remove inner starter fastening stud.

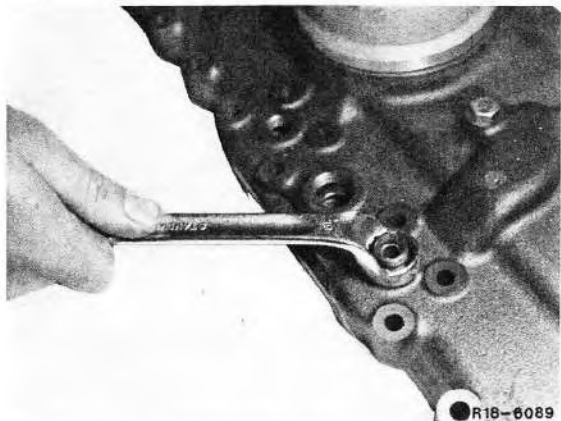
- 6 Detach generator, but do not disconnect wiring.



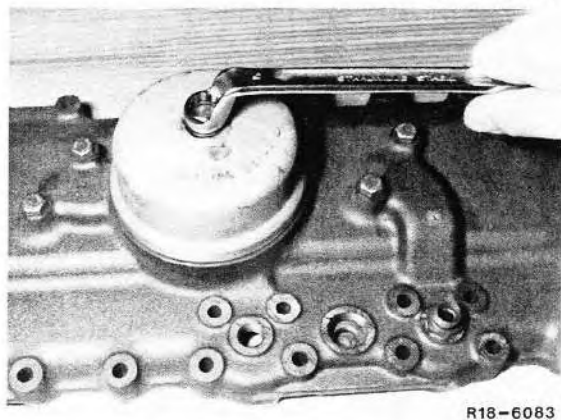
352 A



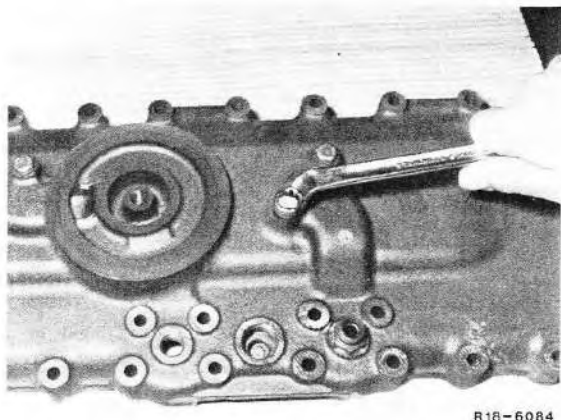
7 Release exterior screw plugs at case cover and two screws at left and right hand of oil pressure relief valve and take case cover with oil cooler.



8 Detach overpressure relief valve and connector to turbocharger.



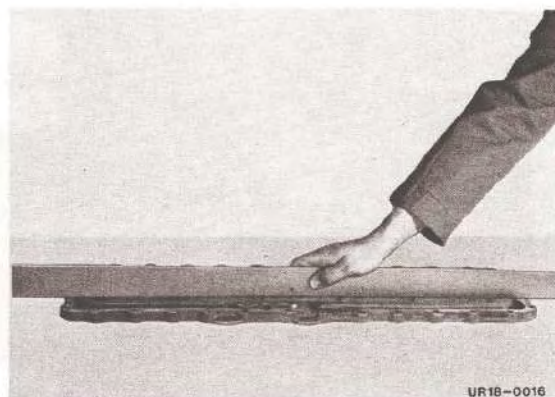
9 Detach filter cover with filter element.



10 Detach oil cooler from case cover.

11 Clean, check and exchange, if necessary, all parts.

12 Check case cover-sealing surface for horizontality.



Installation

From engine No. 528 530

Note: Sequence of operations is identical for all engine models.

1 Attach oil cooler with new gaskets and tighten.



2 Put filter elements into filter cover.

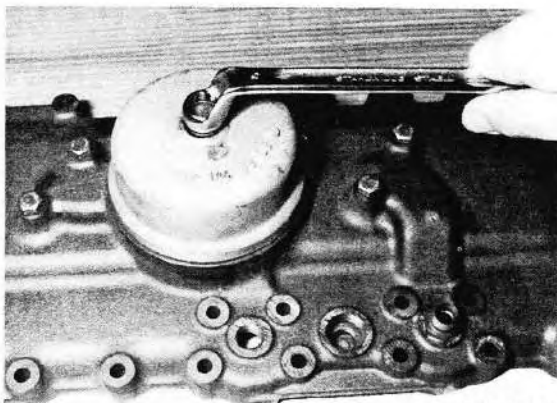
Note: The oil filter downstream of the disk-type oil cooler is maintenance-free. Replace filter element following engine damage, e.g. ingress of water, damage of piston or bearing.



3 Put seal ring in filter cover.

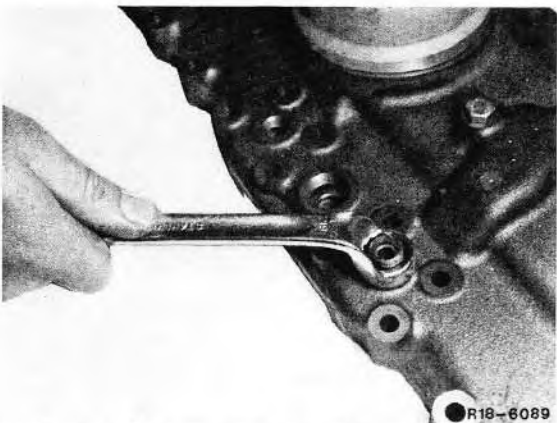


352 A



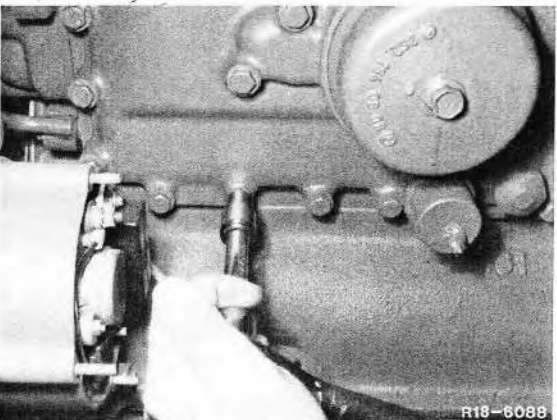
R18-6083

- 4 Attach filter cover to case cover and tighten. Tightening torques refer 1.2/1.



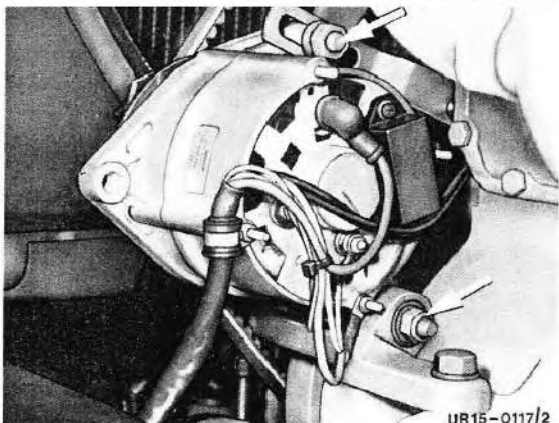
R18-6089

- 5 Attach oil pressure relief valve and connector for oil pressure line to turbocharger. Tightening torques refer 1.2/1.



R18-6088

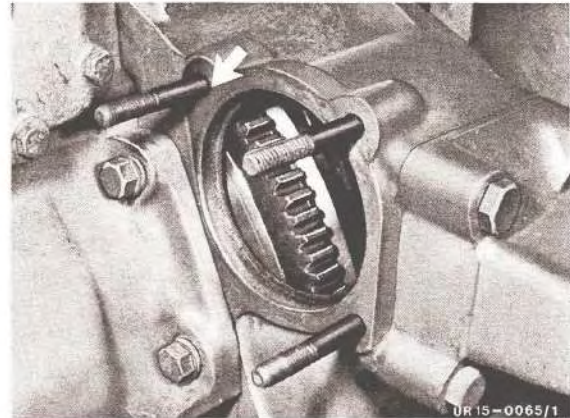
- 6 Attach case cover with new gasket crankcase and tighten all bolts crosswise from outside to inside. Tightening torques refer 1.2/1.



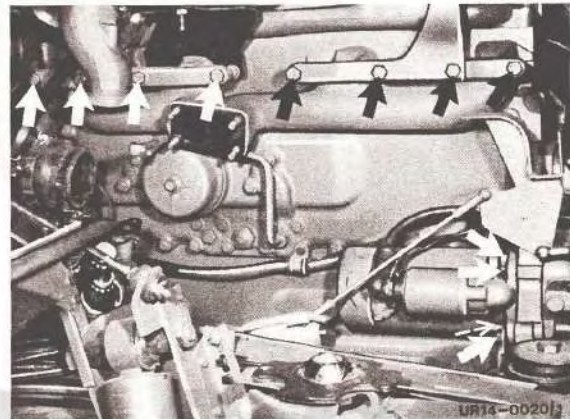
UR15-0117/2

- 7 Attach generator and fit V-bel. Tighten V-belt. Tightening torques refer 13.14-2.6/1.

8 Insert inner stud for starter mounting.
Tightening torque, see 1.2/1.



9 Fit starter with sealant No. 3, screw on nuts with sealant No. 4 and tighten.
Tightening torque, see 1.2/1.



10 Attach and tighten exhaust manifold with new gasket.
Tightening torque, see 1.2/1.

11 Install turbocharger.
Refer to 09.13-2.1/1.

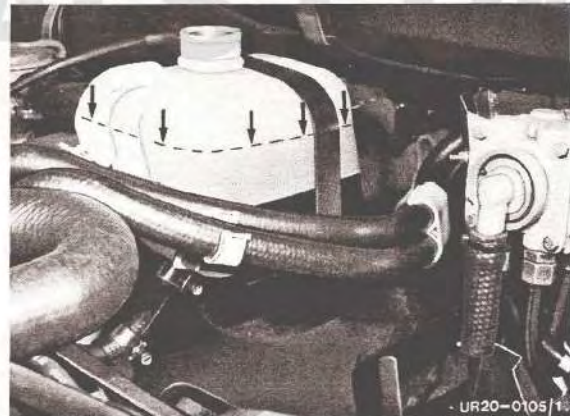
12 Connect battery.

Note: First connect (+), then (-) terminal.

13 Fill coolant.

Refer to filling capacities 01.8-2.1/1.

Note: Reservoir should be 2/3 full.

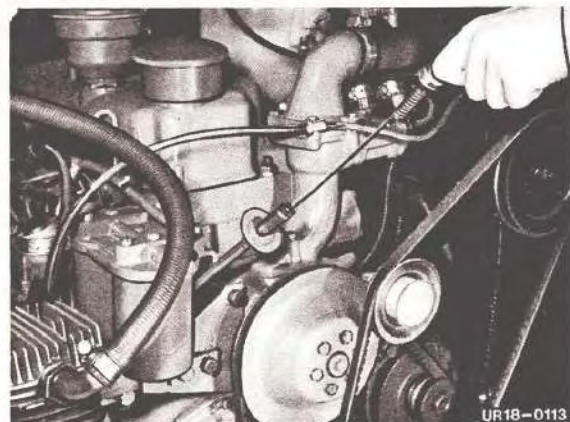


14 Vent engine, referring to 5.1/1.

15 Crank engine with starter, but do not start engine. Engine can be started when pressure gauge shows oil pressure reading.

16 Check oil cooler for leakage, examining engine oil level once again.

17 Lower cab,
referring to 60.5-2.1/1 or 60.5-2.2/1.



Checking

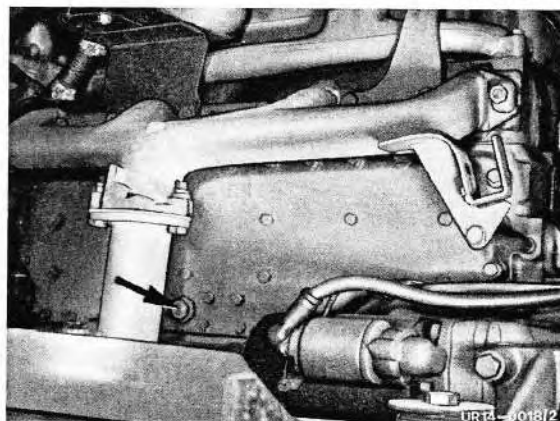
1 Raise hood.



UR88-0030/1

2 Remove screwplug.

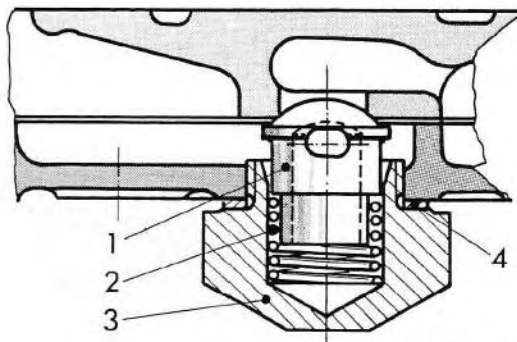
3 Remove sealing ring, spring and valve from oil cooler.



UR14-0018/2

4 Clean all parts, check and replace as necessary

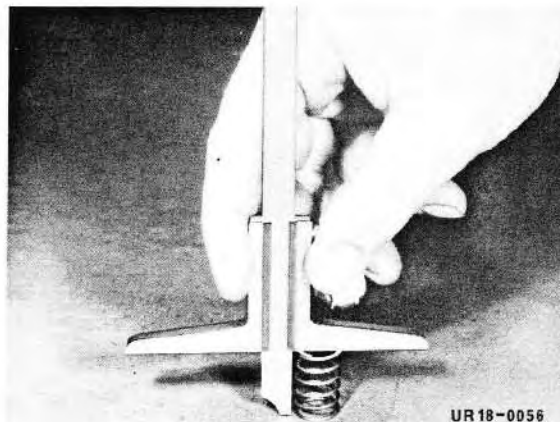
- 1 Valve
- 2 Spring
- 3 Screwplug
- 4 Sealing ring



UZ 18-0028/1

5 Check length of compression spring.
See 1.1/1 for table.

6 Installation takes place in the reverse sequence;
use new sealing ring.



UR 18-0056

Venting with oil filling container (special tool No. 1)

1 Fill special tool No. 1 with approx. 5 liters of engine oil and then close.

Note: Use engine oil according to specifications for service products.

2 Using hand pump incorporated in special tool No. 1, generate approx. 3 bar pressure.

3 Detach cylinder head cover, referring to 01.8–3.1/1.

4 Detach oil pressure pickup and fit special tool No. 1.

5 Open shutoff valve until **bubble-free** oil emerges at rocker arms.

Note: Positive pressure in special tool No. 1 ought not to drop below 1.5 bar.

6 Detach special tool No. 1, and fit oil pressure pickup.

7 Attach cylinder head cover, referring to 01.8–3.1/1.

8 Lift up dipstick locking lever and check oil level, topping up engine oil if necessary.

Venting without oil filling container

1 Fill engine and oil filter housing with oil. Check level on dipstick.
See 01.8–1.3/2 for quantity.

2 Detach cylinder head cover, referring to 01.8–3.1/1.

3 Move injection pump control lever to **stop position**.

4 Crank engine with starter until **bubble-free** oil emerges at rocker arms and pressure gage shows oil pressure reading.

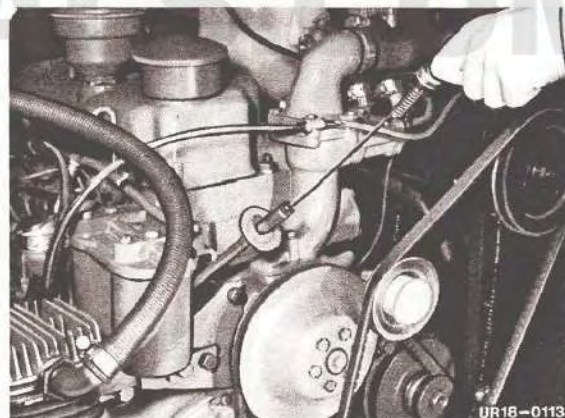
5 Attach cylinder head cover, referring to 01.8–3.1/1.



UR18-0096



UR18-0095/1



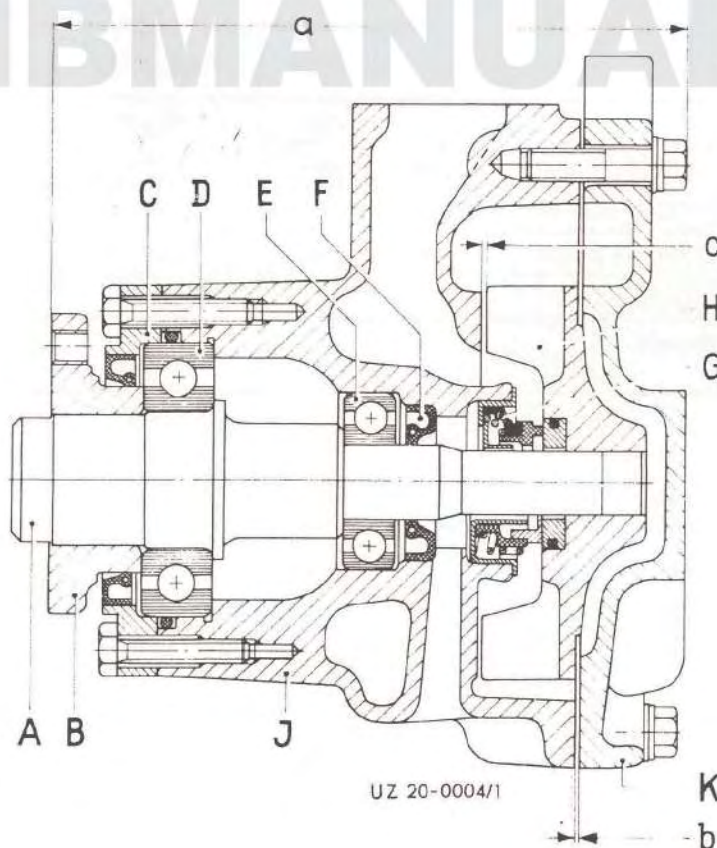
UR18-0113

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5 Checking the cooling system	
Checking the cooling system for leakage	5.1/1
Checking expansion tank cap for satisfactory operation	5.2/1

Technical data
Coolant pump

Pump shaft dia.	at impeller seat	mm	15.039
			15.028
	large bearing	mm	30.009
			29.996
	small bearing	mm	17.008
			16.997
Shaft dia. for hub		mm	29.048
			29.035
Bore dia. in hub		mm	29.021
			29.000
Hub dia. for sealing ring		mm	42.011
			41.995
Shaft dia. for rear sealing ring		mm	17.008
			16.997
Bore dia. in impeller		mm	15.018
			15.000



A Coolant pump shaft

B Hub

C Sealing ring holder

D Large bearing

E Small bearing

F Rear sealing ring

G Mechanical seal

H Impeller

J Coolant pump body

K Coolant pump cover

a Reference size 143.5 ± 0.5 mm

b Press impeller onto shaft until surface "b" is flush with flange surface of coolant pump body.

Max. tolerance ± 0.1 mm

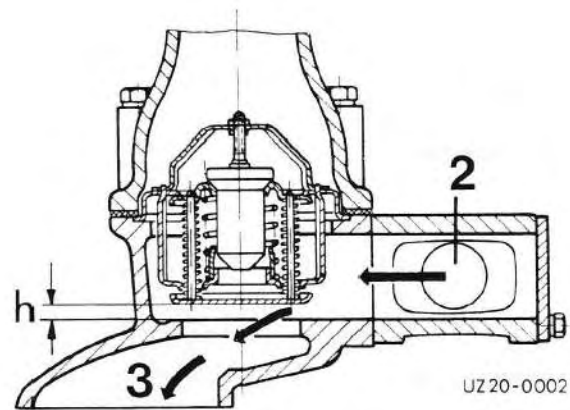
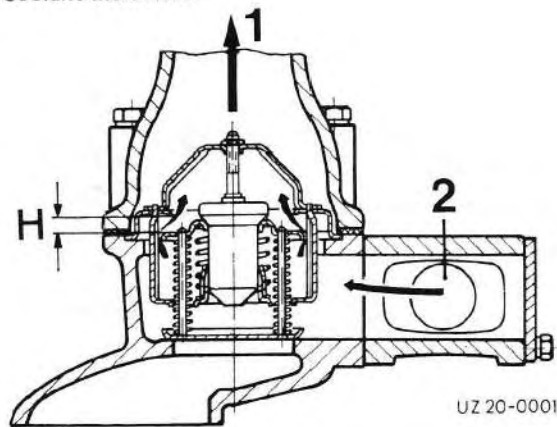
c Clearance between impeller and body

0.6 + 0.5 mm

- 0.3 mm

Coolant pump

Coolant thermostat



Main valve open, bypass line closed

Main valve closed, bypass line open

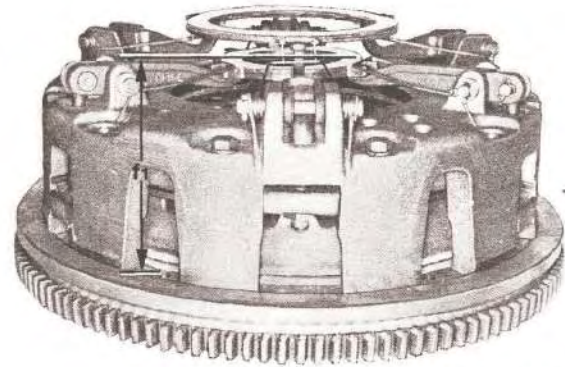
H Main valve stroke
1 to radiator
2 from engine

h Bypass valve stroke
2 from engine
3 to engine via bypass line

Coolant throughput (coolant leakage) of main valve at 1 bar and 20°C in l/min		l/min	0.67 to 1.17
Valve begins to open at		°C	83 ± 2
Main valve	Stroke in mm H	mm	8
	Fully open at	°C	95
Bypass valve	Stroke in mm h	mm	6 to 7.5
	Closed at	°C	92
Max. necessary coolant flow for full opened main valve and 1 bar pressure		l/min	333

Adjustment

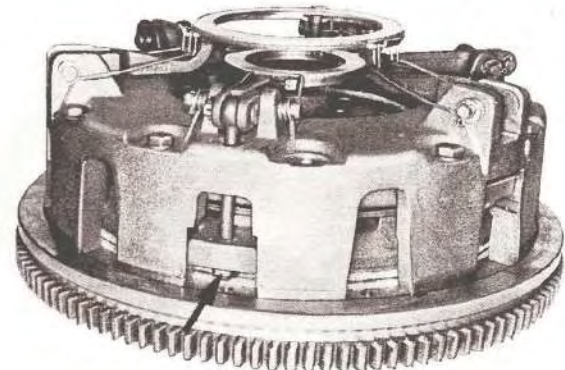
- 1 Check main clutch adjustment, referring to 1.3/1.



UR25-0185/1

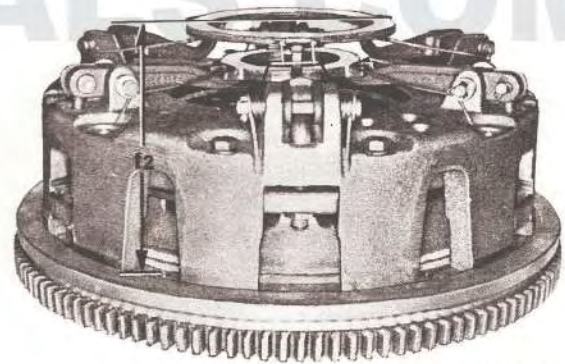
- 2 If necessary, correct adjustment at bolts.

Note: Nuts squeezed at mounting height.



UR25-0186/1

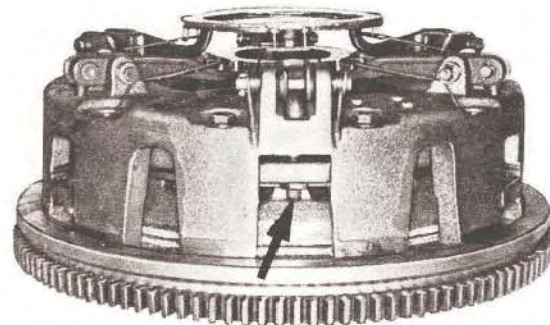
- 3 Check adjustment for PTO shaft clutch, referring to 1.3/1.



UR25-0185/2

- 4 If necessary, correct adjustment at bolts.

Note: Nuts squeezed at mounting height.



UR25-0187/1